KUMARESAN K

B.Tech – Computer Science and Engineering

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SUMMARY

Aspiring Computer Science Engineering student with hands-on experience in full-stack web development using Spring Boot, React, and Node.js. Skilled in building decentralized applications with Solidity and Ethereum, focusing on secure smart contracts and Web3 integration. Passionate about solving real-world problems through clean, efficient code and innovative blockchain-driven solutions.

EDUCATION

Sri Manakula Vinayagar Engineering College, Puducherry **2022 – 2026**

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.5

Petit Seminaire Higher Secondary School

Higher Secondary Education (12th Standard) — May 2022 | Secondary Education (10th Standard) — May 2020

TECHNICAL SKILLS

Programming Languages	Java, JavaScript, Solidity	**Tools**	Git, GitHub, Postman, VS Code, Remix IDE, Foundry
Front-End	HTML, CSS, React, Angular JS	**Database**	MySQL, MongoDB
Back-End	Spring Boot, Node.js, ethers.js	**Cloud Platforms**	AWS (Cloud Foundation)
CS Fundamentals	Data Structures (Java), DBMS, OOP	**Enterprise Systems**	SAP S/4HANA (Financial Accounting), SAP BTP

PROJECTS

PharmaChain Vault **Jan 2026**

Group Project | Role: Blockchain and Backend Developer

- **Idea & Aim:** Created a secure platform to protect and manage sensitive medical formulas using blockchain. Prevent data theft and unauthorized modifications while ensuring transparency and ownership.
- **Description:** Built a blockchain-based platform to store medical formulas as NFTs on Ethereum, integrating AES-256 encryption, IPFS storage, and a React-based UI. Developed Solidity smart contracts with multi-signature access control.
- **Key Characteristics:** Secure encrypted storage with NFT ownership • Multi-signature access control • Transparent and tamper-proof system

Tech Stack: Solidity, React, Node.js, MySQL, ethers.js, Sepolia, WalletConnect, IPFS

Healthr — Decentralized Health Records **June 2025**

Individual Project | Role: Blockchain and Flutter Developer

- **Idea & Aim:** Built a decentralized system to give patients control over their medical records, ensuring secure, transparent, and tamper-proof health data management.
- **Description:** Developed a Flutter-based dApp using Ethereum, integrating WalletConnect for authentication and IPFS for decentralized storage. Implemented Solidity smart contracts for role-based access and deployed on Sepolia testnet.
- **Key Characteristics:** Secure and tamper-proof storage • Role-based access control • Patient-centric data ownership

Tech Stack: Dart, Solidity, Flutter, Ethereum, IPFS, WalletConnect

TSender — ERC20 Airdrop Tool **May 2024**

Individual Project | Role: Blockchain Developer

- **Idea & Aim:** A decentralized ERC20 bulk-transfer application for distributing tokens to multiple wallet addresses in one transaction, simplifying distribution and reducing gas fees compared with separate individual transfers.
- **Description:** Built a responsive dApp with MetaMask connection, network selection, recipient and amount inputs, token allowance checks, automatic approval flow, transaction summary, and transaction status tracking. Supports Ethereum, Arbitrum, Optimism, Base, zkSync Era, Sepolia, and Localhost.
- **Key Characteristics:** Multi-recipient transfers • Multi-chain support • Allowance and approval handling • Gas-efficient bulk transfers • Responsive UI

Tech Stack: HTML5, CSS3, JavaScript, Solidity, ERC20, MetaMask

INTERNSHIPS

Google Android Developer Intern — AICTE (Virtual) **Jul 2025 – Sep 2025**

- Developed Android applications using Kotlin, XML, and Android Studio, focusing on responsive UI and modern app architecture.
- Built feature-based applications including e-commerce modules with user authentication, navigation, and data persistence.
- Implemented MVVM architecture, integrated REST APIs using Retrofit, and managed local storage using Room database.
- Worked with Firebase for backend services including authentication and real-time data handling.
- Gained hands-on experience in app lifecycle, debugging, and performance optimization through project-based learning.

CERTIFICATIONS

- Implementation Consultant — SAP S/4HANA Cloud Public Edition, Financial Accounting
- Solution Architect — SAP BTP
- Programming in Java — NPTEL
- Java Full Stack Development — Wipro
- Database Management System — NPTEL
- Blockchain Development — IBM
- Basic Solidity Smart Contracts — Infosys
- Cloud Foundations — AWS

ACHIEVEMENTS

- Achieved Top 3 recognition for “PharmaChain Vault” in the Best Project competition at Sri Manakula Vinayagar Engineering College in 2026.
- Selected as Semi-Finalist in the Unisys Innovation Program conducted by Unisys.
- Published research paper titled “Pharma-Chain Vault” in IJEDR, Volume 13, Issue 4 (Dec 2025), ISSN 2321-9939.

AREA OF INTEREST

Blockchain Development, Java Full Stack Development, MERN Stack Development, Web3 Development